Department: Computer Science and Engineering (Data Science)

Vision of the Department

"To create a first-class academic department that trains the next generation of students as data scientists to solve the grand challenges and innovate through world-class research to take advantage of these opportunities"

Mission of the Department

- Empower students in a field that has piloted in industrial revolution and the original computing revolution.
- To build strong foundation in Data computation, Intelligent Systems that enables self-development entrepreneurship and Intellectual property.
- Provide an environment for leading edge research that has a strong and rapid impact on the economy and that re-establishes the technological advancement.
- To develop competent and skilled IT professional by imparting global skills and technologies for serving society.

Program Educational Objectives (PEOs)

- To produce educationally favourable environment to learn engineering skills in the domains like Data Analytics, Data Modelling, Data Visualization and other Technologies.
- Graduates will have the capability to apply their knowledge and skills acquired to solve the issues in real world as well in Data Science sectors to develop feasible systems.
- To work efficiently in multi-disciplinary teams to establish good soft skills to prove themselves at IT sector.
- Inculcate lifelong learning abilities and encourage pursuing higher studies and exploring research.

Program Outcomes (POs)

1. **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2. **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

3. **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

4. **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

5. **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

6. **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

7. **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

8. **Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. **Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design

documentation, make effective presentations considering cultural, language, and learning differences

10. **Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. **Life-Long Learning:** Recognize the need for, and have the preparation and ability for

    i) Independent and life-long learning
    ii) Adaptability to new and emerging technologies and
    iii) Critical thinking in the broadest context of technological change.

SRI SIDDHARTHA ACADEMY OF HIGHER EDUCATION TUMAKURU

(Deemed to be University, Accredited **A+** Grade by NAAC, New Delhi.)

SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY- TUMAKURU

(A constituent College of Sri Siddhartha Academy of Higher Education, Tumakuru.)

Syllabus for the Academic Year 2025-26

Scheme of Teaching and Examination-2022 (160 Credits Scheme, NEP)
Outcome-Based Education (OBE) and Choice Based Credit System (CBCS)
(Effective from the academic year 2022-23)

VII Semester B.E **Academic year: 2025-26**

Sl.No.	Course Code		Course Title	Teaching Dept.	Teaching Hrs.					Credits	Examination			
					L	T	P	S	Total		CIE Marks	SEE Marks	Total Marks	Exam Hrs
1	PC	22DS701	Data Analytics	DS	42	-	28	50	120	4	50	50	100	3
2	PC	22DS702	Cloud Computing	DS	42	-	28	50	120	4	50	50	100	3
3	PE	22DS7PE3x	Professional Elective-III	DS	42	-	-	48	90	3	50	50	100	3
4	PE	22DS7PE4x	Professional Elective-IV	DS	42	-	-	48	90	3	50	50	100	3
5	PE	22DS7PE5x	Professional Elective-V	DS	42	-	-	48	90	3	50	50	100	3
6	PW	22DSPW706	Project Work Phase-I	DS	-	-	28	32	60	2	50	-	50	3
7	PC	22DSTS707	Technical Seminar	DS	-	-	-	30	30	1	50	-	50	-
				Total	**210**		**84**	**306**	**600**	**20**	**350**	**250**	**600**	**-**

L: Lecture, T-Tutorial, P-Practical/Drawing, CIE: Continuous Internal Evaluation, SEE: Semester End Examination, S: Self Learning. XX: Branch Code

Credit Distribution: PC: Professional Core-09, PE: Professional Elective-09, PW: Project Work-02, **Total=20 Credits.**

Professional Elective-III	Professional Elective-IV	Professional Elective-V
22DS7PE31: Cryptography & Network Security	22DS7PE41: Artificial Intelligence	22DS7PE51: Information Retrieval System
22DS7PE32: Theory of Computation	22DS7PE42: Block Chain Technology	22DS7PE52: Natural Language Processing with Transformers
22DS7PE33: Robotic Process Automation	22DS7PE43: Web Mining	22DS7PE53: Full Stack Development

Department: CSE (Data Science)	**Semester:**	**7**

Subject: Data Analytics				
Subject Code:	**22DS701**	**L – T – P - S:**	**42-0-28-50**	
			Credits	**4**

Sl. No	Course Objectives
1	To **understand** knowledge of data using location, shape and dispersion methods.
2	To **understand** the importance of different types of data, data visualization.
3	To **gain** knowledge of correlation between data and time series analysis.
4	To **understand** the knowledge of hypothesis testing, data preparation and processing.

Module	Description	Hrs
I	**Introduction:** Difference between descriptive & inferential statistics, data mining and intelligent data analysis. Data and knowledge, Data Understanding: Attribute Understanding, Data Quality. Statistics: Terms and Notations, Descriptive Statistics, Tabular and Graphical Representations (problems on frequency tables) Characteristic Measures for one dimensional data — Locations measures, Dispersion measures, Shape measures, Box plots (numerical problems) Data Visualization –Methods for one and two attributes (histogram, Box plot, stem and leaf plot, Scatter plot). **Text Book1: 1- 1.1.1, 1.1.3 , 4: 4.1, 4.2, 4.3.1,Appendix A – A.1,A.2.2, A.2.3.1-2.3.4,A.2.4,.2.5**	**10**
II	**Characteristics Measures for Multidimensional Data:** Correlation Analysis (problems on Pearson's correlations coefficient, Ranks correlation — Spearman's and Kendall's tau rank correlation coefficient), Visualization Methods for Higher dimensional data (Principle component analysis, Parallel Coordinates, Radar and Star plots). Outlier Detection in single and multidimensional data, a checklist for data understanding. **Linear Regression, Multiple regressions**: multivariate data, partial correlation coefficients, multiple correlations. **Text Book1: 4 : 4.3.2.1, 4.3.2.5, 4.3.2.6, 4.4, 4.5, 4.5.1, 4.5.2, 4.7** **Text Book2 : 17 : 17.1, 17.3, 17.4, 17.5,17.7,18.1.18.2,18.5,18.6**	**8**
III	**Time series Analysis:** Importance of Time series analysis, components of a time series, Trend of time series using method of simple moving average, simple weighted moving average, centred moving average, median smoothening, weighted moving centred average method, method of least square, linear regression method for forecasting, and exponential smoothing(simple and adjusted). Forecasting using seasonal indexing. **Text Book2 : 19: 19.1, 19.2, 19.3,19.7**	**8**

IV	**Testing of Hypothesis**: Introduction to hypothesis testing, Procedure of testing hypothesis, Type I error, Tails of a test, Z test and t test -Lower Tail Test of Population Mean with known Variance, Upper Tail Test of Population Mean with known Variance, Two-Tailed Test of Population Mean with Known Variance, Lower Tail Test of Population Mean with Unknown Variance, Upper Tail Test of Population Mean with Unknown Variance, Two-Tailed Test of Population Mean with Unknown Variance. **Chi-square distribution**: properties, the goodness of fit test, test of independence, Chi-square as test of homogeneity. **Text Book 2 : 13,14:14.1,4.2, 14.3, 14.4**	**8**
V	**Data Preparation: Select Data:** Feature Selection, Dimensionality Reduction Record Selection. **Clean data**: improve data quality, missing values. **Construct data**: Provide operability, assure impartiality (Data transformation). Data Integration: Vertical and Horizontal data integration (different types of joins). **Text Book 1: 6**	**8**

LAB CONTENT

Sl. No	Experiment Description
	Write a R program to perform :
1	Various operations on vectors.
2	Mean median, mode, range, summary, variance, standard deviation operations.
3	Time series Analysis.
4	Various operations on matrices.
5	A Bar chart, scatter plot, box plot and Histogram.
6	Pearson's correlations coefficient.
7	Hypothesis testing.
8	Normal distributions.
9	Binomial l distributions.
10	Linear Regression.

Text Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Guide to Intelligent Data Analysis	Michael R.Berthod, Christian Borgelt, Frank Hoppner	Springer Series, 2019, ISBN: 9781848822597.
2	Business Statistics	GC Beri,	3rd Edition. Tata Mc-GrawHill, 2010.
3	http://www.r-tutor.com/elementary-statistics/hypothesis-testing		

Reference Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Business Analytics: Data Analysis and Decision Making	Christina Albright, Wayne L. Winston	CENGAGE 5th edition.

Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** various locations, shape and dispersion measures and interpret the results for better understanding of data.
CO2	Use various visualization techniques and **interpret** the plots for understanding data.
CO3	**Analyze** different types of correlation among data to interpret the association among data and interpret the outcome.
CO4	**Illustrate** various time series forecasting method and hypothesis testing for prediction and interpret the outcome.
CO5	**Design** and implement end-to-end data preparation and integration pipelines.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	1	2								
CO2	2	2	3	1	2								
CO3	3	3	2	2	2								
CO4	3	3	2	2	3								
CO5			1	2	1								

Department: CSE (Data Science)	**Semester:**	**7**

Subject: Cloud Computing				
Subject Code:	**22DS702**	**L – T – P - S:**	**42-0-28-50**	
			Credits	**4**

Sl. No	Course Objectives
1	**Introducing** the cloud computing models and the business drivers.
2	**Understand** the importance of cloud virtualization
3	**Understand** the design of cloud built-in applications and necessary tools.
4	**Realize** the importance of cloud architecture, cloud programming and cloud security

Module	Description	Hrs
I	**Distributed System Models and Enabling Technologies:** Scalable Computing Over the Internet, Technologies for Network Based Systems, System Models for Distributed and Cloud Computing, Software Environments for Distributed Systems and Clouds, Performance, Security and Energy Efficiency. **Text Book 1: 1.1 to 1.5**	**10**
II	**Virtual Machines and Virtualization of Clusters and Data Centers:** Implementation Levels of Virtualization, Virtualization Structure/Tools and Mechanisms, Virtualization of CPU/Memory and I/O devices, Virtual Clusters and Resource Management, Virtualization for Data Center Automation. **TextBook1: 1:3.1 to 3.5**	**8**
III	**Cloud Platform Architecture over Virtualized Datacenters:** Cloud Computing and Service Models, Data Center Design and Interconnection Networks, Architectural Design of Compute and Storage Clouds, Public Cloud Platforms: GAE, AWS and Azure, Inter-Cloud Resource Management. **TextBook1: 4.1 to 4.5**	**8**
IV	**Cloud Security:** Top concern for cloud users, Risks, Privacy Impact Assessment, Cloud Data Encryption, Security of Database Services, OS security, VM Security, Security Risks Posed by Shared Images and Management OS, XOAR, A Trusted Hypervisor, Mobile Devices and Cloud Security. **TextBook2: 11.1 to 11.3, 11.5 to 11.8, 11.10 to 11.14**	**8**
V	**Cloud Programming and Software Environments:** Features of Cloud and Grid Platforms, Parallel and Distributed Computing Paradigms, Programming Support for Google App Engine, Programming on Amazon AWS and Microsoft, Emerging Cloud Software Environments. **TextBook1: 6.1 to 6.5**	**8**

LAB CONTENT

Sl. No	Experiment Description
1	Creating a Virtual Machine: Configure and deploy a virtual machine with specific CPU and memory requirements in Google Cloud. OR Exploring AWS Cloud Shell and the AWS Cloud9 IDE
2	Getting Started with Cloud Shell and gcloud: Discover the use of gcloud commands to manage Google Cloud resources from Cloud Shell. OR Working with Amazon S3Orchestrating Server less Functions with AWS Step Functions.
3	Cloud Functions: Create and deploy a Cloud Function to automate a specific task based on a Cloud Storage event. OR Working with Amazon Dynamo DB
4	App Engine: Deploy a web application on App Engine with automatic scaling enabled. OR Developing REST APIs with Amazon API Gateway
5	Cloud Storage: Qwikstart: Google Cloud Storage provides scalable and secure object storage for managing data, accessible via the Cloud Console or gsutil CLI. OR Creating Lambda Functions Using the AWS SDK for Python
6	Cloud SQL for MySQL: Discover how Google Cloud SQL for MySQL provide automated management and high availability for MySQL databases? OR Migrating a Web Application to Docker Containers
7	Cloud Pub/Sub: Experiment how Google Cloud Pub/Sub facilitate real-time messaging and communication between distributed applications. OR Caching Application Data with ElastiCache, Caching with Amazon CloudFronT, Caching Strategies
8	Multiple VPC Networks: Explore benefits of using multiple VPC networks in Google Cloud for organizing and isolating resources. OR Implementing CloudFront for Caching and Application Security
9	Cloud Monitoring: Discover how Cloud Monitoring help in tracking and analyzing the performance and health of cloud resources? OR Orchestrating Serverless Functions with AWS Step Functions
10	Kubernetes Engine: Qwik Start: Deploy a containerized application to a Kubernetes Engine cluster. OR Automating Application Deployment Using a CI/CD Pipeline

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Distributed and Cloud Computing	Kai Hwang, Geoffrey C Fox, and Jack J Dongarra	Morgan Kaufmann, Elsevier 2012
2	Cloud Computing Theory and Practice	Dan C. Marinescu,	Morgan Kaufmann, 2nd Edition, Elsevier 2018
3	Google Cloud Teaching Resources LMS	-	for Lab component
4	AWS Cloud Developing – AWS Academy Courses	-	for Lab component

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Mastering Cloud Computing	Rajkumar Buyya, Christian Vecchiola, and Thamrai Selvi	McGrawHill Education, 1st Edition, 2017

Course Outcomes:

Course outcome	Descriptions
CO1	**Describe** various cloud computing platforms and service providers.
CO2	**Illustrate** the significance of various types of virtualization.
CO3	**Recognize** the service models and public cloud platforms used in cloud computing applications.
CO4	**Analyze** the significance of security measures in cloud computing environments.
CO5	**Demonstrate** cloud applications in various fields

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3										
CO2				2									
CO3			1		1								
CO4						3	2						
CO5	3	3	3										

Department: CSE(Data Science)			**Semester:**	**7**
Subject: Cryptography and Network Security				
Subject Code:	**22DS7PE31**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Understand** security mechanism with encryption and decryption ciphers.
2	**Illustrate** various security algorithms.
3	To **analyze** key management and internet security techniques.
4	Able to **analyze** and illustrate Network Security.

Module	Description	Hrs
I	**Computer and Network Security Concepts:** Computer security, Information security, and Network security, OSI Security architecture, Security attacks, Security services, Security mechanisms, A Model for Network Security **Symmetric Ciphers: Classical Encryption Techniques:** Symmetric Cipher Model, Substitution Techniques, Transposition Techniques, Steganography. **Text Book 1: Chapter: 1 and 3**	**08**
II	**Block Cipher and the Data Encryption Standard:** Block Cipher principles, The Data Encryption Standard, The Strength of DES, Electronic Code Book, Cipher Block Chaining Mode, Cipher Feedback Mode. **Number Theory and Public Key Cryptosystem: Number Theory:** Prime Numbers, Format's and Euler's Theorems, Testing for Primality. **Public-Key Cryptography and RSA**: Principles of Public- Key Cryptosystems, The RSA Algorithm. Diffie-Hellman Key Exchange. **Text Book 1: Chapter 4, 2, 9, 10**	**09**
III	**Cryptographic Data Integrity Algorithms: Cryptographic Hash Functions:** Two simple hash Functions, Secure Hash Algorithm. **Message Authentication Codes:** Message Authentication Requirements, Message Authentication Functions, Digital Signatures. **Key Management and Distribution:** Symmetric Key distribution using symmetric encryption, Symmetric Key distribution using Asymmetric encryption, Distribution of public keys, X.509 Certificates, Kerberos. **Text Book 1: Chapter: 11, 12, 13, 14, 15**	**09**
IV	**Transport level security:** Web Security considerations, Layer and Transport Layer Security. **Electronic Mail Security:** Internet Mail Architecture, Email threats and Comprehensive Email Security. **IP Security:** IP Security Overview, IP Security Policy. **Text Book 1: Chapter: 17, 19,20**	**08**
V	**Network Security, Firewalls, and Virtual Private Networks (VPN):** Introduction, Introduction to TCP/IP, Firewalls, Virtual Private Networks (VPN), Intrusion. **Text Book 2: Chapter 9**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Cryptography and Network Security	William Stallings	Seventh Edition, Prentice Hall of India, 2016.
2	Cryptography and Network Security	AtulKahate	Tata McGrawHill, 4th Edition 2019

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Network Security: Private Communication in a Public World	Charlie Kaufman, Radia Perlman, Mike Speciner	Second Edition, Pearson,Education Asia.
2	Cryptography, Network Security and Cyber Laws	Bernard Menezes	Cengage Learning, 2018

Course Outcomes:

Course outcome	Descriptions
CO1	**Understand** and **apply** fundamental concepts of computer and network security, and symmetric encryption techniques.
CO2	**Analyze** block cipher principles, DES, and public-key cryptography concepts.
CO3	**Apply** cryptographic techniques for data integrity, authentication, digital signatures, and key management.
CO4	**Evaluate** transport-level, email, and IP security mechanisms.
CO5	**Analyze** network security components including TCP/IP security, firewalls, VPNs

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1		2								
CO2	3	3	2		2								
CO3	3	2	2		2								
CO4	2	2	3		2								
CO5	3	3	2		2								

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Theory of Computation				
Subject Code:	**22DS7PE32**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Understand** core concepts in Automata and Theory of Computation
2	**Identify** different Formal Language Classes and their Relationships
3	**Prove** or disprove theorems in Automata Theory using their properties.
4	**Compare** finite automata, Push Down Automata and Turing Machines as Mathematical models of computation.

Module	Description	Hrs
I	**Introduction to Finite Automata:** Why study Automata Theory, The Central Concepts of automata theory: Alphabets, Strings, Languages, Definition of DFA. How a DFA processes strings, Simpler notations for DFA's, Extending the transition function to strings, The language of a DFA, Nondeterministic Finite Automata (NFA),An informal view of NFA, Definition of NFA, An Extended Transition Function, The language of NFA, Equivalence of DFA and NFA(Using Lazy Evaluation method), Finite Automata with Epsilon-transitions. **Text Book 1: Chapter: 1.1,1.5,2.2,2.3.2.5.**	**09**
II	**Regular Expressions and Languages, Properties of Regular:** Regular Expressions, Converting DFA Regular Expressions by state eliminating method, Converting Regular Expressions to Finite Automata, Applications of Regular Expressions: Lexical Analysis, Finding Patterns in Text; Proving languages not to be Regular, Closure properties of Regular Languages, Equivalence and Minimization of Automata, Testing equivalence of states, Testing equivalence of Regular Languages, Minimization of DFA's. **Text Book 1: Chapter :3.1,3.2,3.3,4.2,4.4.**	**09**
III	**Context-Free Grammars and Languages:** Context-Free Grammars, An informal example, Definition of Context-Free Grammars, Derivations using a Grammar, The language of a grammar, Sentential forms, Parse Trees, Constructing Parse Trees, The yield of a Parse tree, Inferences, Derivations and Parse Trees Applications of Context-Free Grammars, Ambiguity in Grammars and Languages, Ambiguous Grammars, Removing ambiguity from grammars, Leftmost derivations as a way to express ambiguity, Inherent ambiguity. **Text Book 1: Chapter : 5.1,5.2,5.3,5.4**	**08**
IV	**Pushdown Automata:** Definition of the Pushdown Automata, Informal Definition, Formal Definition, A Graphical Notation for PDA's, Instantaneous Descriptions of PDA The Languages of a PDA. Acceptance by Final State, Acceptance by Empty Stack, Equivalence of PDA's and CFG's. From Grammars to PDA's. From PDA's to Grammars, Definition of Deterministic PDA . **Text Book 2: Chapter: 6.1,6.2,6.3,6.4.**	**08**

V	**Properties of Context-Free Languages**,: Normal Forms for CFG's. Eliminating Useless Symbols, Computing the Generating and Reachable Symbols, Eliminating € Productions, Unit Productions. **The Turing Machine**: Notation for the Turing Machine, Instantaneous Descriptions for Turing Machines, Transition Diagrams for Turing Machines, The Language of a Turing Machine. **Undecidability:** Enumerating the Binary Strings, Codes for Turing Machines, The Diagonalization Language, Recursive Languages, Complements of Recursive and RE languages, **Text Book 1: Chapter :7.1,8.2,9.1,9.2**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Introduction to Automata Theory Languages and Computation	John E Hopocroft Rajeev Motwani, Jeffery D Ullman	3rd Edition,Pearson Education Limited,2014 ISBN-13:9781292039053
2	Theory of Computer Science Automats Languages and Computation	KLP Mishra, N Chandrasekaran	3rd Edition PH, 2016 ISBN :9788120329183

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Finite Automata and Formal Languages: A Simple Approach	A. M. Padma Reddy, Reddy AM Padma	ISBN:9788131760475, 8131760472 Publisher:Pearson Education India 2011

Course Outcomes:

Course outcome	Descriptions
CO1	**Understanding** of the core concepts in automata theory and Theory of Computation
CO2	**Apply** the knowledge to translate between different models of Computation (e.g., Deterministic and Non-deterministic and Software models).
CO3	**Design** Grammars and Automata (recognizers) for different language classes and become knowledgeable about restricted models of Computation (Regular, Context Free) and their relative powers.
CO4	**Apply** formal reasoning and reduction of a problem to a formal model.
CO5	**Develop** skills to emphasis on semantic precision and conciseness.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	2	3						2		
CO2	3	3	3	3	2						2		
CO3	2	3	3	3	3								
CO4		3	3	3	3						2		
CO5		3	2	3							2		

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Robotic Process Automation				
Subject Code:	**22DS7PE33**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	**Course Objectives**
1	**Understand** fundamental concepts of automation.
2	Learn and **Understand** UI Automation activities.
3	Learn and **automate** Mail Automation and Word Automation activities.
4	Understand **and analyze** File Automation and Presentation Automation activities.

Module	**Description**	**Hrs**
I	**Robotic Process Automation: Overview:** Return on Investment (ROI), Automation Types, UiPath StudioX. **Common Concepts:** Notebook, Activity Inputs, Activity Outputs, Common Properties, Common Activities, Write Line: Message Box, Input Dialog. Modify Text., Text to Left/Right, Delay, if, Switch., Repeat Number Of Times. , Skip Current, Exit Loop, Get Username/Password, Get Orchestrator Asset, Save For Later, Wait for Download, Group. **Text Book 1:Chapter 1 , Chapter 3**	**08**
II	**UI Automation:** Sample Overview, Activities Reference, Use Application/Browser, Go To URL, Navigate Browser, Highlight, Take Screenshot, Check App State, Click, Type Into, Select Item, Check/Uncheck, Get Text, Get Attribute, Extract Table Data, Hover, Keyboard Shortcuts, Get Active Window, Maximize Window, Minimize Window, Hide Window, Restore Window, Move Window. App/Web Recorder. **Text Book 1:Chapter 4**	**09**
III	**Mail Automation:** Sample Overview: Desktop Outlook Setup, File System Structure, Activities Reference: Use Desktop Outlook App, Use Gmail, For Each Email, Mark Email As Read/Unread, Forward Email, Save Email Attachments, Save Email, Send Email, Send Calendar Invite, Move Email, Reply to Email, Archive Email, Delete Email. **Word Automation:** Sample Overview: Word Setup, File System Structure, Activities Reference: Use Word File..., Save Document As, Read Text, Set Bookmark Contant, Replace Text in Document, Append Text, Insert DataTable in Document, Replace Picture,Add Picture, Save Document as PDF. **Text Book 1:Chapter 5 , Chapter 6**	**09**

IV	**Excel Automation:** Sample Overview, Activities Reference, Use Excel File, Insert Sheet, Rename Sheet, Duplicate Sheet, Delete Sheet, For Each Excel Sheet, Insert Column,Text To Columns..,Delete Column, Insert Rows, Delete Rows,Find First/Last Data Row, For Each Excel Row, WriteCell, Create Pivot Table, Format as Table, Change Pivot Data Source, Refresh Pivot Table, Append Range, Copy Range, Sort Range, Clear Sheet/Range/Table, Auto Fill, Fill Range, Write Range, Read Cell Formula.., Read Cell Value, Format Cells, Export to CSV.., Save Excel File, Save Excel File As...,Save Excel File As PDF, VLookup, Filter, Run Spreadsheet Macro. **Text Book 1:Chapter 7**	**08**
V	**File Automation:** Sample Overview. Activities Reference: Get Folder Info, Folder Exists, Create Folder, Delete Folder, Copy Folder., Move Folder, For Each File In Folder, Compress/Zip Files, Extract/Unzip Files, Get File Info, File Exists, Create File, Delete File, Copy File, Move File, Write Text File, Append Line.., Read Text File. **Presentation Automation:** Sample Overview, File System Structure., Activities Reference, Use PowerPoint Presentation, Copy Paste Slide, Delete Slide, Add New Slide, Replace Text in Presentation, Add Text to Slide., Add Data Table to Slide, Add Image/Video to Slide, Add File to Slide., Run Presentation Macro, Save PowerPoint File As., Save Presentation as PDF. **Text Book 1:Chapter 9, Chapter 10**	**08**

Text Books:

Sl.No	Text Book title	Author	Volume and Year of Edition
1	Robotic Process Automation using UiPath StudioX: A Citizen Developer's Guide to Hyper automation,	Adeel Javed, Anum Sundrani, Nadia Malik, Sidney Madison Prescott,	1st Edition, Apress, April 2021

Reference Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	The Robotic Process Automation Handbook : A Guide to Implementing RPA Systems,	Tom Taulli ,	ISBN- 13 (electronic): 978-1-4842-5729-6, Publisher : Apress, 2020
2	Learning Robotic Process Automation, Publisher: Packt	Alok Mani Tripathi,	Publishing Release Date: March 2018 ISBN: 9781788470940

Course Outcomes:

Course outcome	Descriptions
CO1	**Understand** the fundamentals of Robotic Process Automation, ROI, types of automation, and apply basic UiPath StudioX concepts.
CO2	**Design** and implement UI automation workflows using UiPath StudioX.
CO3	**Automate** email communication and document processing tasks using UiPath activities.
CO4	**Apply** advanced automation techniques in Excel using UiPath.
CO5	**Develop** automation solutions for file system operations and PowerPoint using UiPath.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2		3								
CO2	2	3	3		3								
CO3	2	2	3		3								
CO4	2	3	3		3								
CO5	2	2	3		3								

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Artificial Intelligence				
Subject Code:	**22DS7PE41**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Understand** and apply intelligent agents and searching algorithms to solve problems.
2	**Understand** knowledge based agents and solve knowledge based engineering problems using first order logic.
3	**Compare** propositional and first order inference, forward and backward chaining.
4	**Understand** stochastic approaches to solve Planning and uncertainty problems.

Module	Description	Hrs
I	What is AI? Acting humanly and thinking humanly, thinking rationally and acting rationally, Intelligent Agents: Agents and Environments, Good Behavior: The concept of Rationality: Rationality, Omniscience, Learning and autonomy, The nature of Environments: specifying the task environment, properties of task environments, The structure of Agents: Agent Programs, simple reflex agents, Model-based reflex agents, Goal-based agents, Utility-based agents, Learning agents, How the components of agents programme work. Solving problems by Searching: problem-solving agents, well-defined problems and solutions, Example problems. Searching for Solutions: infrastructure for search algorithms, measuring problem-solving performance **Text Book 1: Chapter 1:1.1, Chapter 2, Chapter 3: 3.1-3.3**	**08**
II	**Uninformed Search strategies**: BFS, uninform-cost search, DFS, depth-limited search, iterative deepening depth-first search, bidirectional search, comparing uniformed search strategies, Informed search strategies: Greedy best-first search, A* search, Memory-bounded heuristic search, learning to search better. Adversarial search: Games, Optimal Decisions in Games- The minimax algorithm, Optimal decisions in multiplayer games, Alpha–Beta Pruning. **Text Book 1: Chapter 3.4, 3.5, chapter 5: 5.1, 5.2, 5.3**	**08**
III	**Constraint satisfaction problems**: Example problem: Map coloring, Example problem: Job-shop scheduling, Variations on the CSP formalism, constraint propagation: Inference in csps:Node consistency, Arc consistency, Path consistency, K-consistency, Global constraints, Backtracking search for CSPs; Variable and value ordering, Interleaving search and inference, Intelligent backtracking: Looking backward. Knowledge-based agents; The wumpus world as an example world, Logic; propositional logic: a very simple logic A simple knowledge base, A simple inference procedure, propositional theorem proving: Inference and proofs, Proof by resolution, Horn clauses and definite clauses, Forward and backward chaining. First order logic: Syntax and semantics of first- order logic, Using first-order logic, Knowledge engineering infirst-orderlogic. **Text Book 1: Chapter 6:6.1-6.3, Chapter 7:7.1-7.5, Chapter 8: 8.2-8.4**	**08**

IV	**Inference In First-Order Logic**: Propositional vs. First-order Inference, Unification and Lifting, Forward chaining: First-order definite clauses, A simple forward-chaining algorithm, Efficient forward chaining, backward chaining: A backward-chaining algorithm, Resolution: Conjunctive normal form for first-order logic, The resolution inference rule, Example proofs. Knowledge Representation: Ontological Engineering, Categories and Objects, Events, Mental events and Mental Objects. Reasoning systems for Categories, Reasoning with default information. **Text Book 1: Chapter 9, Chapter 10 (12 in 3rd edition)**	**08**
V	**Classical Planning**: Definition of Classical Planning, Algorithms For Planning as State-Space Search, Planning Graphs, other Classical Planning Approaches, Analysis of Planning Approaches QuantifyingUncertainty: Acting under Uncertainty, Basic Probability Notation, Inference using Full Joint Distributions, Independence, Bayes' Rule and its Use, The Wumpus World Revisited . **Text Book 1: Chapter 11 (10 in 3rd edition), 12 (13 in 3rd edition)**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Artificial Intelligence - A Modern Approach	Stuart Russel Peter Norvig	Pearson Education 3rd Edition 2013/4th Edition 2021

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Artificial Intelligence	Elaine Rich	Tata McGraw Hill 3rd Edition, 2019
2	Introduction to Artificial Intelligence	Wolfgang Ertel	Springer 2nd Edition, 2017

Course Outcomes

Course outcome	Descriptions
CO1	**Understand** artificial intelligent agents for simple applications.
CO2	**Apply** searching algorithms to develop artificial intelligence agents.
CO3	**Develop** knowledge base sentences with propositional logic and first order logic.
CO4	**Design** AI representational system for reasoning about simple applications
CO5	**Apply and analyze** the stochastic approaches to Planning and uncertainty.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1		2										2	
CO2													
CO3			2									2	
CO4			2									2	
CO5			2									2	

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Block Chain Technology				
Subject Code:	**22DS7PE42**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	To **Understand** Blockchain terminologies with its applications and design
2	**Describe** the working principles of blockchain and **analyze** the methodologies used in the Bitcoin protocol.
3	**Demonstrate** knowledge of Ethereum architecture including wallets, nodes, and smart contracts.
4	**Design** blockchain-based applications using Hyperledger and **explain** the smart contract development and deployment lifecycle.

Module	Description	Hrs
I	**Blockchain:** Distributed systems, CAP theorem, Byzantine Generals problems, Consensus,Introduction to blockchain, various technical definitions of blockchains, Generic elements of a blockchain, The history of blockchain, Features of a blockchain, Applications of blockchain technology, Tiers of blockchain technology, Types of Blockchain, CAP theorem and blockchain, Benefits and limitations of blockchain. **Text Book 1:Chapter 1**	**08**
II	**Decentralization**: Decentralization using Blockchain, Methods of decentralization, routes to decentralization, Blockchain and full echo system decentralization, Smart Contract, Decentralize organizations,Decentralized autonomous organization, Decentralized autonomous corporation, decentralized autonomous societies, Decentralized applications, Platforms for decentralization Cryptographic primitives, Symmetric cryptography, DES, AES **Text Book 1: chapter 2 ,chapter 3**	**09**
III	**Bitcoin:** Bitcoin definition, Transactions, The transaction life cycle, The transaction structure, Types of transaction, The structure of a block , The structure of a block header, The genesis block, The bitcoin network, Wallets, Bitcoin payments, **Alternative Coins**-Theoretical foundations, Bitcoin limitations. **Text Book 1 : chapter 4,5**	**08**
IV	**Smart Contracts and Ethereum 101**: Smart Contracts: Definition, Ricardian contracts. Ethereum 101: Introduction, Ethereum blockchain, Elements of the Ethereum blockchain, Precompiled contracts, Accounts, Block, Ether, Messages, The Ethereum network, Applications developed on Ethereum **Text Book 1: chapter 6,7**	**08**
V	Hyperledger, Hyperledger as a protocol, Fabric, Alternative Blockchains: Blockchains- Kadena**,** Blockchain-Outside of Currencies: Internet of Things, Government, Health, Finance, Media **Text Book 1: chapter 9,10,11**	**09**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Mastering Blockchain	Imran Bashir	Packt, 2017

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Blockchain Basics	D. Drescher Apress	2017
2	Mastering Bitcoin	Andreas M Antonopoulos	O'Reilly Media publications, First Edition

Course Outcomes:

Course outcome	Descriptions
CO1	**Explain** fundamental concepts of blockchain and various types and applications of blockchain.
CO2	**Analyze** decentralization methods, smart contracts, and cryptographic foundations.
CO3	**Describe** the Bitcoin architecture, transaction lifecycle, and block structure, and examine limitations and use of altcoins.
CO4	**Demonstrate** understanding of smart contracts and evaluate the core components of the Ethereum blockchain
CO5	**Apply** knowledge of Hyperledger Fabric and explore alternative blockchains

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2					1						2	
CO2	2	3				1					2	2	
CO3		2			1	1	1				2		
CO4	2				1				1		2	2	
CO5			2	1		2	1		1		1		

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Web Mining				
Subject Code:	**22DS7PE43**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Understand** the concepts of web search and retrieval of information
2	**Compare** the different mining methods based on algorithm
3	**Analyze** the structure of Data on Web
4	**Understand** the web mining tools and its applications

Module	Description	Hrs
I	**Information Retrieval and Web Search**: Basic Concepts of Information Retrieval, Information Retrieval Model- Boolean Model, Vector Space Model, Statistical Language Model, Relevance Feedback, Evaluation Measures, Text and Web Page Pre-Processing, Inverted Index and Its Compression, Latent Semantic Indexing, Web Search, Web Spamming. **Text Book 1: 6.1,6.2,6.3,6.4,6.5,6.6,6.7,6.8,6.9,6.10**	**08**
II	**Social Network Analysis** : Social Network Analysis-Centrality, Prestige, Co- Citation and Bibliographic Coupling, PageRank- PageRank Algorithm, Strengths and Weaknesses of PageRank, Timed PageRank and Regency Search, HITS- HITS Algorithm, Finding Other Eigenvectors, Relationships with Co-Citation and Bibliographic coupling, Community Discovery **Text Book 1: 7.1,7.2,7.3,7.4,7.5**	**09**
III	**Web Crawling** : A Basic Crawler Algorithm- Breadth-First Crawlers, Preferential Crawler, Implementation Issues- Fetching, Parsing, Stopword Removal and Stemming, Link Extraction and Canonicalization, Spider Traps , Page Repository, Concurrency, Universal Crawlers, Focused Crawlers, Evaluation, Crawler Ethics and Conflicts, **Text Book 1:8.1,8.2,8.3,8.4,8.5,8.6,8,7.**	**08**
IV	**Structured Data Extraction**: Wrapper Generation Preliminaries, Wrapper Induction, Instance-Based Wrapper Learning, Automatic Wrapper Generation: Problems, String Matching and Tree Matching, Multiple Alignment, Building DOM Trees, Extraction Based on a Single List Page, Extraction Based on Multiple Pages. **Text Book 1 :9.1,9.2,9.3,9.4,9.5,9.6,9.7,9.8,9.9,9.10**	**09**
V	**Information Integration**: Introduction to Schema Matching, Pre-Processing for Schema Matching, Schema-Level - Linguistic Approaches, Constraint Based Approaches, Domain and Instance-Level Matching, Combining Similarities, 1:m Match, Some Other Issues, Integration of Web Query Interfaces, Constructing a Unified Global Query Interface **Text Book 1: 10.1,10.2,10.3,10.4,10.5,10.6,10.7,10.8,10.9**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Web Data Mining	Bing Liu	Springer 2nd edition,2011

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Web Mining: Applications and Techniques	Dr. Anthony Scime	2nd edition 2004

Course Outcomes:

Course outcome	Descriptions
CO1	**Understand** the different web search and retrieval of information
CO2	**Analyse** web contents and Social Networks to provide analytics with relevant visualization tools.
CO3	**Compare and analyze** the performance of web mining.
CO4	**Identify** different Integration Issues and applications.
CO5	**Analyze** different web Queries and Interfaces.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2									2	2	2	
CO2	2	2										2	
CO3	2				2						2	2	
CO4	2			2							2	2	
CO5	2	2		2								2	

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Information Retrieval System				
Subject Code:	**22DS7PE51**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Understand** the fundamentals of Information Retrieval systems and its role in modern web- based systems.
2	**Examine** and Explore various search interfaces and learn how user interaction influences IR system design.
3	**Apply** dictionary-based indexing techniques and **implement** methods such as tokenization, stemming, and spelling correction..
4	**Analyse** evaluation metrics, feedback mechanisms, and Examine the architecture of web crawlers

Module	Description	Hrs
I	**Introduction**: Information retrieval, IR problem, IR System, The web. **User interfaces for search:** Introduction, How people search, Search interfaces today, Visualization on search interfaces, Design and evaluation of search interfaces. **Text Book1: Chapter 1:1.1 to 1.4, Chapter 2:2.1 to 2.5**	**08**
II	**Dictionaries and tolerant retrieval:** Search structures for dictionaries, Wildcard queries: General wildcard queries , k-gram indexes for wildcard queries , Spelling correction: Implementing spelling correction, Forms of spelling correction, Edit distance, k-gram indexes for spelling correction, Context sensitive spelling correction, Phonetic correction. **Index construction :**Hardware basics, Blocked sort-based indexing, Single-pass in-memory indexing, Distributed indexing, Dynamic indexing ,Other types of indexes **Text Book 2: chapter 3 ,chapter 4**	**08**
III	**Modeling:** IR models, Classic information retrieval, Alternative set theoretic models, Alternative algebraic models, Alternative probabilistic models, Other models. **Text Book 1: Chapter 3:3.1 to 3.6**	**08**
IV	**Retrieval Evaluation:** Retrieval metrics, Reference Collections, User-based evaluation **Relevance feedback and Query expansion:** A framework for feedback methods, Explicit relevance feedback, Explicit feedback through clicks, Implicit feedback through local analysis, Implicit feedback through global analysis **Documents - Languages and Properties:** Metadata, Document formats, Text properties, Document preprocessing, Organizing documents, Text compression **Text Book 1: Chapter 4:4.3 to 4.5, Chapter 5:5.2 to 5.6, Chapter 6:6.2 to 6.3, 6.5 to 6.8**	**09**

V	**Web search basics : Background and history** Web characteristics: The web graph, Spam, Advertising as the economic model, The search user experience: User query needs, Index size and estimation, Near- duplicates and shingling **Web crawling and indexes:** Overview: Features a crawler must provide, Features a crawler should provide, Crawling: Crawler architecture: DNS resolution, The URL frontier, Distributing indexes, Connectivity servers **Link analysis:**The Web as a graph: Anchor text and the web graph, PageRank: Markov chain, The PageRank computation, Topic-specific PageRank , Hubs and Authorities: Choosing the subset of the Web. **Text Book 2: chapter 19 ,chapter 20, chapter 21**	**09**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Modern Information Retrieval	Ricardo Baeza-Yates and Berthier Ribeiro-Neto	2nd Edition, Pearson 2011
2	Introduction to Information Retrieval	C. Manning, P. Raghavan, and H. Schutze	Cambridge University Press, 1st Edition, 2009

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Information Retrieval – Algorithms and Heuristics	David A. Grossman, OphirFrieder	Springer, 2nd Edition(Distributed by Universal Press), 2004
2	Information Retrieval Implementing and Evaluating Search Engines	Stefan Buttcher Charles L. A. Clarke Gordon V. Cormack	1st Edition, February 2016

Course Outcomes:

Course outcome	Descriptions
CO1	**Explain** IR system fundamentals and analyze the role of user interfaces in search.
CO2	**Apply** dictionary indexing and spelling correction techniques for tolerant retrieval.
CO3	**Evaluate** classical and advanced IR models for retrieval effectiveness.
CO4	**Assess** IR system performance and analyze feedback and preprocessing methods.
CO5	**Explain** web IR challenges and demonstrate crawling and link analysis algorithms.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1			1								
CO2	2	2	2	2	2						2		
CO3	1	3									2		
CO4		2		2									
CO5		2	2	2			1				1		

Department: CSE (Data Science)		**Semester:**	**7**
Subject: Natural Language Processing with Transformers			
Subject Code:	**22DS7PE52**	**L – T – P - S:**	**42-0-0-48**
			Credits: **3**

Sl. No	Course Objectives
1	**Understand** the importance of natural language modeling.
2	**Identify** the Applications of natural language processing.
3	**Analyze** spelling, error detection and correction methods and parsing techniques in NLP.
4	**Analyze** the different approaches of Machine Translation and transformers

Module	Description	Hrs
I	**Introduction:** What is Natural Language Processing? Origins of NLP, Language and Knowledge, The Challenges of NLP, Language and Grammar, Processing Indian Languages, NLP Applications. **Language Modeling:** Statistical Language Model - N-gram model (unigram, bigram), Paninion Framework, Karaka theory. **Text Book 1: 1.1,1.2,1.3,1.4,1.5,1.6,1.7,2.3**	**08**
II	**Word Level Analysis:** Regular Expressions, Finite-State Automata, Morphological Parsing, Spelling Error Detection and Correction, Words and Word Classes, Part-of Speech Tagging. **Syntactic Analysis:** Context-Free Grammar, Constituency, To Parsing. **Semantic Analysis:** Introduction, Meaning Representation, Lexical Semantics, Ambiguity, Word Sense Disambiguation. **Text Book 1: 3.2,3.3,3.4,3.5,3.6,3.7,4.2,4.3,4.4,5.1,5.2,5.3,5.4,5.5**	**09**
III	**Information Retrieval**: Design Features of Information Retrieval Systems, Information Retrieval Models, Classical Information Retrieval Models, Non-classical models of IR, Alternative Models of Information Retrieval, Evaluation of the IR System. **Lexical Resources:** WorldNet, Frame Net, Stemmers, Parts-of-Speech Tagger, Research Corpora. **Text Book 1: 9.2,9.3,9.4,9.5,9.6,9.7,12.2,12.3,12.4,12.5,12.6.**	**08**
IV	**Machine Translation**: Introduction, Problems in Machine Translation, Characteristics of Indian Languages, Machine Translation Approaches, Direct Machine Translation, Rule-based Machine Translation, Corpus-based Machine Translation, Semantic or Knowledge-based MT systems, Translation involving Indian Languages. **Text Book 1: 8.1,8.2,8.3,8.4,8.5,8.6,8.7,8.8**	**08**
V	**Transformer Anatomy:** Transformer Anatomy, The Encoder, The Decoder, Meet the Transformers- The Transformer Tree of Life, The Encoder Branch, The Decoder Branch, The Encoder-Decoder Branch. **Multilingual Named Entity Recognition:** The Dataset, Multilingual Transformers, A Closer Look at Tokenization- The Tokenize Pipeline, The Sentence Piece Tokenizer, Transformers for Named Entity Recognition. **Text Book 2: Ch. 3, Ch. 4.**	**09**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	"Natural Language Processing and Information Retrieval"	Tanveer Siddiqui, U.S. Tiwary,	Oxford University Press.
2	Natural Language Processing with Transformers	Lewis Tunstall, Leandro von Werra & Thomas Wolf	O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472 Feb-2022

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Text Data with Mach"Natural Language Processing Recipes — Unlocking Machine Learning and Deep Learning using Python	Akshay Kulkarni, Adarsha Shivananda,	Apress, 2019.
2	"Understanding Natural Language Processing – Machine Learning and Deep Learning Perspectives	T V Geetha	Pearson, 2024.
3	"Information Storage and Retrieval systems"	Gerald J. Kowalski and Mark.T. Maybury Kluwer	Academic Publishers.
4	"Speech and Language Processing, An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition".	Daniel Jurafsky, James H. Martin	Pearson Education, 2023.

Course Outcomes:

Course outcome	Descriptions
CO1	**Understand** the fundamental concept of NLP.
CO2	**Analysis** the different Models syntactically and semantically.
CO3	**Analyse** the concepts of information retrieval, lexical semantics and lexical dictionaries.
CO4	**Design** the challenges and techniques used in transformers anatomy and entity recognition
CO5	**Analyze** the transformer anatomy Encoder-Decoder.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3								2		
CO2	3	2	3										
CO3	3	2		2									
CO4	3	2		2							2		
CO5	3	2	2										

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Full Stack Development				
Subject Code:	**22DS7PE53**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	To **understand** the essential JavaScript concepts for web development.
2	To **style** Web applications using bootstrap and utilize React JS to build front end User Interface.
3	To **understand** the usage of API's to create web applications using Express JS.
4	To **store** and **model** data in a no sql database.

Module	Description	Hrs
I	Basic JavaScript Instructions, Statements, Comments, Variables, Data Types, Arrays, Strings, Functions, Methods & Objects, Decisions & Loops. **Text Book 1: Chapter 2, 3, 4**	**08**
II	Document Object Model: DOM Manipulation, Selecting Elements, Working with DOM Nodes, Updating Element Content & Attributes, Events, Different Types of Events, How to Bind an Event to an Element, Event Delegation, Event Listeners. Form enhancement and validation. **Text Book 1: Chapter: 5, 6, 13**	**10**
III	Introduction to MERN: MERN components, Server less Hello world. **React Components:** Issue Tracker, React Classes, Composing Components, Passing Data Using Properties, Passing Data Using Children, Dynamic Composition. **Text Book 2: Chapter 1, 2, 3**	**08**
IV	**React State:** Initial State, Async State Initialization, Updating State, Lifting State Up, Event Handling, Stateless Components, Designing Components, State vs. Props, Component Hierarchy, Communication, Stateless Components. **Express**, REST API, GraphQL, Field Specification, Graph Based, Single Endpoint, Strongly Typed, Introspection, Libraries, The About API GraphQL Schema File, The List API, List API Integration, Custom Scalar types, The Create API, Create API Integration, Query Variables, Input Validations,Displaying Errors. **Text Book 2: Chapter 4, 5**	**08**
V	**MongoDB:** Basics, Documents, Collections, Databases, Query Language, Installation, The Mongo Shell, MongoDB CRUD Operations, Create, Read, Projection, Update, Delete, Aggregate, MongoDB Node.js Driver, Schema Initialization, Reading from MongoDB, Writing to MongoDB. **Modularization and Web pack, Back**-End Modules Front-End Modules and Web pack Transform and, Bundle, Libraries Bundle ,Hot Module Replacement, Debugging Define Plugin: Build Configuration, Production Optimization. **Text Book 2: Chapter 6, 8**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	"JavaScript & jQuery: Interactive Front-End Web Development".	Jon Duckett,	Wiley, 2014.
2	Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React,and Node.	Vasan Subramanian	Apress, 2019.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Mastering Full Stack React Web Development	Tomasz Dyl , Kamil Przeorski , Maciej Czarnecki	Paperback – April 28, 2017

Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** JavaScript to build dynamic and interactive Web projects.
CO2	**Implement** user interface components for JavaScript-based Web using React.JS.
CO3	**Apply** Express/Node to build web applications on the server side.
CO4	**Develop** data model in an open source nosql database
CO5	**Demonstrate** modularization and packing of the front-end modules.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3		3				2	2			
CO2	3	2	3		3				3	2			
CO3	3	2	3	2	3	1			2	2			
CO4	3	2	3		3				3	2			
CO5	3	2	3		3								

Department: CSE (Data Science)	**Semester:**	**7**

Subject: Project Work Phase – I				
Subject Code:	**22DSPW706**		**L – T – P - S:**	**0-0-28-32**
			Credit	**2**

Sl. No	Experiment Description
1	The students are divided into batches (same discipline or interdisciplinary), with each batch consisting of a minimum of 2 students and a maximum of 4 students (including bright and slow learners) for Major Projects,
2	Each batch must submit three different synopses in a particular domain of their interest within one week from the commencement of the semester.
3	Faculties are allocated to the project batches as guides, considering their area of specialization.
4	Students MUST get approval of the project title from the respective guide and the Department Project Evaluation Committee (DPEC).
5	After finalization of the project title and guide allotment, students should give a presentation to their respective guides.
6	Interdisciplinary projects must have a guide from each of the participating departments.
7	The students should maintain a project summary report consisting of the day-to-day work carried out by them, monitored weekly by the guide.
8	The Project work phase-I done by students, is evaluated for 50 CIE Marks **No SEE**

Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** theoretical knowledge and engineering principles to design and implement a practical solution for a real-world problem.
CO2	Effectively **plan, manage, and execute** a multi-disciplinary project, adhering to timelines and resource constraints.
CO3	**Analyze, interpret, evaluate** experimental results and identify the areas for improvement.
CO4	**Communicate** complex technical information effectively through reports, presentations, and demonstrations.
CO5	**Demonstrate** ethical considerations, professional conduct, and an understanding of intellectual property rights through their team work.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	2	1		1	1	1	1		
CO2	2	2	3	1	2			1	2	2	2		
CO3	3	3	2	3	2			2	2	2			
CO4			2					3	2	3			
CO5								3	3	2	1		

Department: CSE (Data Science)			**Semester:**	**7**
Subject: Technical Seminar				
Subject Code:	**22DSTS707**	**L – T – P - S:**	**0-0-0-30**	
			Credits:	**1**

Guidelines for preparing Technical Seminar
1. Selection of Topic/Domain: Each student shall identify a topic/domain of current cutting edge technology in the field of Computer Science and Engineering. Student is also allowed to choose the recent research paper/article of the standard journal. **2. Approval of the Topic:** After selecting the topic or research paper, get approval by the concerned faculty. **3. Collect and acquire thorough knowledge:** Students are required to collect sufficient information (past and recent updates) about the topic they have chosen and allowed to read the related research papers and refer the books. **4. Final Report and Presentation:** Students are required to prepare a final report of the collected information about the topic chosen and prepare a Power point presentation under the guidance of the concerned faculty.

Course Outcomes:

Course outcome	Descriptions
CO1	**Identify** and **explore** recent advancements or emerging technologies relevant to the engineering domain.
CO2	**Analyze** and **synthesize** information from diverse technical literature sources.
CO3	**Develop** effective communication skills through preparation, delivery of technical presentations and dissertations
CO4	**Demonstrate** the ability to work independently and exhibit self-learning capabilities.
CO5	**Exhibit** professional ethics and understand the societal impact of modern technologies discussed in seminars.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2			2	2							
CO2	2	3		2	2								
CO3					2		1		3				
CO4									3		2		
CO5						2	2	3					

SRI SIDDHARTHA ACADEMY OF HIGHER EDUCATION TUMAKURU
(Deemed to be University, Accredited **A+** Grade by NAAC, New Delhi.)
SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY- TUMAKURU
(A constituent College of Sri Siddhartha Academy of Higher Education, Tumakuru)
Syllabus for the Academic Year 2025-26

Scheme of Teaching and Examination-2022 (160 Credits Scheme, NEP) Outcome-Based Education (OBE) and Choice Based Credit System (CBCS)
(Effective from the academic year 2022-23)

VIII Semester B.E **Academic year: 2025-26**

Sl. No.	Course Code		Course Title	Teaching Dept.	Teaching Hrs.				Total	Credits	Examination			
					L	T	P	S			CIE Marks	SEE Marks	Total Marks	Exam Hrs
1	PE	22DS8PE1X	Professional Elective-VI	DS	42	-	-	48	90	3	50	50	100	3
2	PE	22DS8PE2x	Professional Elective-VII	DS	42	-	-	48	90	3	50	50	100	3
3	PW	22DSPW803	Project Work Phase-2	DS	-	-	126	114	240	8	50	50	100	3
4	PC	22DSIN804	Internship (Mandatory Non-Credit Course)		-	-	-	-	-	-	50	-	50	-
				Total	**84**	**0**	**126**	**210**	**420**	**14**	**200**	**150**	**350**	**-**

L: Lecture, T-Tutorial, P-Practical/Drawing, CIE: Continuous Internal Evaluation, SEE: Semester End Examination , S:Self Learning. XX: Branch Code

Credit Distribution : PE: Professional Elective-06, PW: Project Work-08, **Total=14 Credits**.

HS: Humanity & Social Sciences-11,BS: Basic Science-22,ES: Engineering Science-23,PC: Professional Core-67, PE: Professional Elective-21,OE: Open Elective- 06, Project Work/Seminar/Internship =10,Total Credits=160.

Professional Elective-VI
- 22DS8PE11: Augmented Reality
- 22DS8PE12: Business Intelligence and Analytics
- 22DS8PE13: Quantum computing

Professional Elective-VII
- 22DS8PE21: Social media networks
- 22DS8PE22: Data security and privacy
- 22DS8PE23: Optimization Techniques

Department: CSE (Data Science)			Semester:	8
Subject: Augmented Reality				
Subject Code:	22DS8PE11	L – T – P - S:	42-0-0-48	
			Credits:	3

Sl. No	Course Objectives
1	**Understanding** basics of Augmented Reality (AR) technology.
2	**Analyze** the hardware requirement of AR.
3	**Explore** tracking and sensor technologies employed in AR systems.
4	**Acquire** knowledge of Computer Vision for Augmented Reality, Calibration and Registration.

Module	Description	Hrs
I	**Introduction to Augmented Reality (A.R)** Introduction to Augmented Reality (AR): Definition and Scope, A Brief History of Augmented Reality, Examples, Related Fields. **Displays:** Multimodal displays, Visual Perception Requirements and Characteristics, Spatial Display Model, visual display. **Text Book 1: Chapter 1, Chapter 2**	**09**
II	**Tracking & Sensors** - Tracking, Calibration, and Registration, coordinate systems, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors, Optical Tracking, Sensor Fusion. **Text Book 1: Chapter 3**	**09**
III	**Computer Vision for Augmented Reality** - Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, incremental tracking Simultaneous Localization and Mapping, Outdoor Tracking. **Text Book 1: Chapter 4**	**08**
IV	**Calibration and Registration** **Camera Calibration-**Internal Camera Parameters, Correcting Lens Distortion, Display Calibration-Single Point Active Alignment Method, Head-Mounted Display Calibration Using a Pointing Device, Hand–Eye Calibration, **Registration-** Geometric Measurement Distortions, Error Propagation, Latency, Filtering and Prediction. **Text Book 1: Chapter 5**	**08**
V	**Visual Coherence** Registration, occlusion, photometric registration, common illumination, diminished reality, camera simulation, stylized augmented reality. **Situated visualization** Challenges, visualization registration, annotations and labeling, X-ray visualization. **Text Book 1: Chapter 6, Chapter 7**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Augmented Reality: Principles & Practice	Schmalstieg Hollerer	Pearson Education India; First edition (12 October 2016),ISBN-10: 9332578494

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Theory and applications of marker-based augmented reality	Sanni Siltanen	Julkaisija — Utgivare Publisher. 2012. ISBN 978-951-38-7449-0
2	Designing for Mixed Reality	Kharis O'Connell	O'Reilly Media, Inc., 2016, ISBN: 9781491962381

Course Outcomes:

Course outcome	Descriptions
CO1	**Describe** the basic concepts of Augmented Reality (AR) technology.
CO2	**Analyse** the hardware requirement of AR.
CO3	**Demonstrate** the tracking and sensor technologies employed in AR systems.
CO4	**Understand** the various concepts of Computer Vision for Augmented Reality, Calibration and Registration.
CO5	**Identify** visual coherence techniques and visualization methods used in AR.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1											
CO2	3	2	2		1								
CO3	3	1	2		2								
CO4	3	2	2		2								
CO5	3	2	1		2				1				

Department: CSE(Data Science)			**Semester:**	**8**
Subject: Business Intelligence and Analytics				
Subject Code:	**22DS8PE12**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Understand** the importance of business intelligence, analytics and Decision Support system.
2	**Acquire** knowledge of Decision making.
3	**Illustrate** Decision Support systems and model based decision making.
4	**Explore** various concepts of Automated Decision Systems and Expert Systems.

Module	Description	Hrs
I	**An Overview of Business Intelligence, Analytics, and Decision Support:** Information Systems Support for Decision Making, An Early Framework for Computerized Decision Support, The Concept of Decision Support Systems, A Framework for Business Intelligence, Business Analytics Overview, Brief Introduction to Big Data Analytics. **Text Book 1: Chapter 1:1.1, 1.4 to 1.9**	**09**
II	**Decision Making:** Introduction and Definitions, Phases of the Decision, Making Process, The Intelligence Phase, Design Phase, Choice Phase, Implementation Phase, Decision Support Systems Capabilities, Decision Support Systems Classification, Decision Support Systems Components. **Text Book 1: Chapter 2:2.2 to 2.7, 2.9 to 2.11**	**09**
III	**Business Reporting, Visual Analytics, and Business Performance Management** Business Reporting Definitions and Concepts, Data and Information Visualization, Different Types of Charts and Graphs, The Emergence of Data Visualization and Visual Analytics, Performance Dashboards, Business Performance Management **Text Book 1: Chapter:4.1 to 4.7**	**08**
IV	**Model-Based Decision Making: Optimization and Multi-Criteria Systems** Decision Support Systems modeling, Structure of mathematical models for decision support, Certainty, Uncertainty, and Risk, Decision modeling with spreadsheets, Mathematical programming optimization, Decision Analysis with Decision Tables and Decision Trees, Multi-Criteria Decision Making With Pairwise Comparisons. **Text Book 1: Chapter 9:9.2 to 9.6, 9.8 to 9.9**	**08**
V	**Automated Decision Systems and Expert Systems:** Automated Decision Systems, The Artificial Intelligence field, Basic concepts of Expert Systems, Applications of Expert Systems, Structure of Expert Systems, Knowledge Engineering, and Development of Expert Systems. **Text Book 1: Chapter 11:11.2 to 11.7, 11.9**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Business Intelligence and Analytics: System for Decision Support	Ramesh Sharda, Dursun Delen, EfraimTurban, J.E.Aronson,Ting-Peng Liang, David King	10th Edition, Pearson Global Edition, 2013

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Data Analytics: The Ultimate Beginner's Guide to Data Analytics	Edward Mize.	Paperback – 12 November 2017 by Edward Mize

Course Outcomes:

Course outcome	Descriptions
CO1	**Able** to gain the knowledge of Business Intelligence, Analytics and Decision Support.
CO2	**Explain** the technologies for Decision making.
CO3	**Demonstrate** the Decision Support systems and model based decision making.
CO4	**Analyse** the concepts of Automated Decision Systems and Expert Systems.
CO5	**Interpret** business reports and visual analytics to support performance management.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2			2								
CO2	3	3	1		2								
CO3	3	2	2		3								
CO4	3	2	2		3								
CO5	3	2	2		3				2	2			

Department: CSE (Data Science)			Semester:	8
Subject: Quantum Computing				
Subject Code:	22DS8PE13	L – T – P - S:	42-0-0-48	
			Credits:	3

Sl. No	Course Objectives
1	To **know** the background of classical computing and quantum computing.
2	To **learn** the fundamental concepts behind quantum computation.
3	To **study** the details of quantum mechanics and the relation to Computer Science.
4	To **gain** knowledge about the basic hardware and mathematical models of quantum computation.

Module	Description	Hrs
I	**Fundamental Concepts** Global Perspectives, Quantum Bits, Quantum Computation, Quantum Algorithms, Experimental Quantum Information Processing, Quantum Information. **Text Book 1:Chapter 1:1.1 to 1.6**	**09**
II	**Introduction To Quantum Mechanics** **Quantum Mechanics:** Linear Algebra, Postulates of Quantum Mechanics, Application: Super dense Coding, Density Operator, The Schmidt Decomposition and Purifications, EPR and the Bell Inequality. **Text Book 1:Chapter 2:2.1 to 2.6**	**09**
III	**Quantum Computation** Quantum Circuits: Quantum Algorithms, Single qubit operations, Controlled operations Measurement Universal Quantum Gates, Quantum Circuit Model of Computation, Simulation of quantum systems **Text Book 1:Chapter 4:4.1 to 4.7**	**08**
IV	**Quantum Information** Quantum Noise and Quantum Operations: Classical Noise and Markov processes, Quantum Operations, Examples of quantum noise and quantum operations, Applications of quantum operations, Distance Measures for Quantum Information. **Text Book 1:Chapter 8:8.1 to 8.4,chapter 9:9.1-9.2**	**08**
V	**Quantum Information Theory** Distinguishing quantum states and the accessible information, Data Compression, Classical Information Over Noisy Quantum Channels, Quantum Information Over Noisy Quantum Channels, Entanglement as a physical resource, Quantum Cryptography. **Text Book 1:Chapter 12:12.1 to 12.4,12.6**	**08**

Text Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Quantum Computation and Quantum Information.	Michael A. Nielsen, Isaac L. Chuang	Tenth Edition, Cambridge University Press, 2010.

Reference Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Data Analytics: The Ultimate Beginner's Guide to Data Analytics	Edward Mize.	Paperback – 12 November 2017 by Edward Mize
2	Quantum Computing Since Democritus.	Scott Aaronson	Cambridge University Press, 2013

Course Outcomes:

Course outcome	Descriptions
CO1	**Understand** the basics of quantum computing.
CO2	**Understand** the background of Quantum Mechanics.
CO3	**Analyse** the computation models.
CO4	**Model** the circuits using quantum computation.
CO5	**Apply** quantum information theory in areas like data compression and cryptography.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1											
CO2	3	2											
CO3	3	2	2		2								
CO4	3	2	2		3								
CO5	3	2	2		3					1			

Department: CSE(Data Science)			Semester:	8
Subject: Social Media Networks				
Subject Code:	22DS8PE21	L – T – P - S:	42-0-0-48	
			Credits:	3

Sl. No	Course Objectives
1	**Describe** the key components and structure of social media networks.
2	**Model** and **visualize** social networks using appropriate tools and representations.
3	**Apply** basic algorithms and techniques for detecting and **analyzing** decentralization in social networks.
4	**Examine** patterns of human behavior within social networks and assess their influence on network dynamics.

Module	Description	Hrs
I	**Introduction to social media networks:** World wide web, web 1.0, web 2.0, web 3.0, social media, core characteristics of social media, types of social media, social networking sites, using facebook for business purposes, content communities, using youtube for business purposes, blogs, microblogging, online collaborative projects Common network terms, common social media network types, common network terminologies, network analytics tools **Text Book1:Chapter 2, Chapter 4**	**08**
II	Visualizing Online Social Networks , A Taxonomy of Visualizations, Graph Representation , Centrality, Clustering , Node-Edge Diagrams, Visualizing Social Networks with Matrix, Based Representations, Node-Link Diagrams , Hybrid Representations , Modelling and aggregating social network data **Text Book 2:Chapter 27,chapter 28**	**09**
III	**Detecting communities in social networks** Definition of community , Evaluating communities — Methods for community detection , Tools for detecting communities **Decentralized online social networks** Introduction , Challenges for DOSN , The Case for Decentralizing OSNs , General Purpose DOSNs , Specialized Application Centric DOSNs , Social Distributed Systems , Delay,Tolerant DOSN. **Text Book 2:Chapter 12,chapter 17**	**08**
IV	**Understanding and predicting human behaviour for social communities**: User data management, Inference and Distribution, Enabling new human experiences — The Technologies.**Managing Trust in Online Social Networks**: Trust in online environment, Trust models based on subjective logic , Trust network analysis , Trust transitivity analysis , Combining trust and reputation — Trust derivation based on trust comparisons. **Text Book 2:Chapter 20,chapter 22**	**08**

V	Evolution in Social Networks — Framework , Tracing Smoothly Evolving Communities , Models and Algorithms for Social Influence Analysis , Influence Related Statistics , Social Similarity and Influence , Influence Maximization in Viral Marketing, Algorithms and Systems for Expert Location in Social Networks, Expert Location without Graph Constraints , Expert Location with Score Propagation , Expert Team Formation. **Text Book 3:Chapter 6(2,5), Chapter 7,chapter 8(1-6)**	09

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Seven Layers of Social Media Analytics Mining Business Insights from social media Text,Actions, Networks,	Gohar F. Khan	2015
2	Handbook of Social Network Technologies and Applications	Borko Furht	1st Edition, 2010.
3	Social Network Data Analytics	Charu C.Aggarwal	2014

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Web Mining and Social Networking — Techniques and applications	Guandong Xu ,Yanchun Zhang and Lin Li	First Edition Springer, 2011.
2	Social Networks and the Semantic Web	Peter Mika	1st ed 2007

Course Outcomes:

Course outcome	Descriptions
CO1	**Describe** the evolution of the web and **classify** various types of social media platforms and their business applications.
CO2	**Model** and **visualize** social networks using graph-based representations and visualization techniques.
CO3	**Apply** algorithms to **detect** communities and **analyze** decentralization in social networks.
CO4	**Examine** user behaviour and **evaluate** trust mechanisms in online social communities.
CO5	**Analyse** social influence, **evaluate** network evolution, and **design** strategies for expert identification in social media environments.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2											
CO2	3	2	2										
CO3	3	3	2								2		
CO4	2	3		2		2	2				2		
CO5	3	3		2		1	1				1		

Department: CSE(Data Science)			**Semester:**	**8**
Subject: Data Security and Privacy				
Subject Code:	**22DS8PE22**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Demonstrate** foundational concepts of information security.
2	**Understand** security management principles, policies, and planning frameworks in organizational contexts.
3	**Explore** technical mechanisms for protecting information systems.
4	I**mplement** security applications in the field of Information technology.

Module	Description	Hrs
I	**Introduction to Information Security**, What Is Security?, Components Of An Information System, Security And The Organization, Information Security: Is It An Art Or A Science?, The Need for Information Security, Information Security Threats And Attacks, The 12 Categories Of Threats **Textbook 1: 1,2**	**08**
II	**Information Security Management:** Introduction to the Management of Information Security, Information Security Planning And Governance, Information Security Policy, Standards,And Practices, Security Education, Training And Awareness Program, Information Security Blueprint,Models And Frameworks, Introduction To Information Security Implementation, The Systems Development Life Cycle, Information Security Project Management, Technical Aspects Of Implementation **Textbook 1: 3,11**	**08**
III	**Security Technology: Access Controls, Firewalls, and VPNs:** Introduction To Access Controls, Firewall Technologies, Protecting Remote Connections, Introduction To Intrusion Detection And Prevention Systems, Honeypots, Honeynets, And Padded Cell Systems, Scanning And Analysis Tools **Textbook 1: chapter 8,9**	**08**
IV	**An Introduction to privacy preserving data mining** Privacy-Preserving Data Mining Algorithms, The Randomization Method, Group Based Anonymization. **Text Book 2: 2.1 to 3.7**	**09**
V	**Distributed Privacy** Distributed Privacy-Preserving Data Mining, Privacy-Preservation of Application Results, Limitations of Privacy: The Curse of Dimensionality, Applications of Privacy-Preserving Data Mining **Textbook 2: 4.1-7.4**	**09**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Principles of Information Security	Michael E. Whitman and Herbert J. Mattord	7th edition
2	Privacy Preserving Data Mining: Models and Algorithms	C. Aggarwal, Philip S Yu, Kluwer	2008

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Cryptography and Network Security	Atul Kahate	4th Edition

Course Outcomes:

Course outcome	Descriptions
CO1	**Identify** and **classify** various types of information security threats .
CO2	**Analyze** and **design** an information security policy and governance structure aligned with organizational goals.
CO3	**Apply** security technologies such as firewalls, VPNs, and intrusion detection systems to **implement** secure IT infrastructure.
CO4	**Evaluate** key management and distribution techniques for securing symmetric and asymmetric encryption systems.
CO5	**Implement** privacy-preserving data mining methods and assess their effectiveness in distributed environments.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2										1		
CO2	2	3	2	2							1		
CO3			2								2		
CO4	1		1			2	2				2		
CO5	1					2	1				2		

Department: CSE (Data Science)			**Semester:**	**8**
Subject: Optimization Techniques				
Subject Code:	**22DS8PE23**	**L – T – P - S:**	**42-0-0-48**	
			Credits:	**3**

Sl. No	Course Objectives
1	**Investigate** on the usage of optimization techniques.
2	To **solve** problems in linear programming.
3	**Implementation** of different direct and indirect methods of optimization.
4	**Analysis** of modern algorithms based on optimization using suitable dataset.

Module	Description	Hrs
I	Introduction, Historical Development, Engineering Applications of Optimization Statement of an Optimization Problem, Design Vector ,Design Constraints , Constraint Surface, Objective Function , Objective Function Surfaces ,Classification of Optimization Problems, Optimization Techniques, Single - Variable Optimization , Multivariable Optimization with no Constraints, Multivariable Optimization with equality constraints , Multivariable Optimization with inequality constraints. **Text Book 1: chapter 1:1.1-1.6 , chapter 2:2.1-2.5**	**08**
II	Linear Programming Problem, Formulation of Linear Programming Model, Important Terminology, Graphical Method, Corner Point Method, Simplex Method ,Condition for Optimal Solution ,Algorithm for Simplex Method ,To Solve an LPP using Artificial Variable Techniques, Algorithm for the Dual Simplex Method **Text Book 2: chapter 3, chapter 4:4.1-4.4,chapter 5:5.3-5.7**	**09**
III	Formulation of Transportation Problem , Initial Basic Feasible Solution of a Transportation Problem , North-West Corner Rule , Least Cost Entry Method (LCM) , Vogel's Approximation Method (VAM), Formulation of Assignment Problem using Hungarian Algorithm , Maximal Assignment Problem , Unbalanced Assignment Problem , Assignment Problem with Restrictions , Travelling Salesman Problem **Text Book 2: chapter 8:8.1-8.6 , chapter 9**	**08**
IV	Unrestricted Search, Dichotomous Search, Quadratic Interpolation Method, Newton's method and quasi-Newton methods ,Unconstrained Optimization Techniques**,** Direct Search Methods: Grid search methods, Powell's method, ,Indirect methods**,** Steepest descent, Newton's method, constrained Optimization Techniques, Direct Random search method, Sequential linear programming , Indirect methods **,** Interior penalty function method, Exterior penalty function method **TextBook1:chapter5:5.3,5.5,5.10,5.12,chapter:6:6.3,6.6,6.9,6.11,chapter 7:7.3,7.5,7.13,7.15**	**09**
V	Genetic Algorithms , Simulated Annealing, Particle Swarm Optimization, Ant Colony Optimization, Optimization of Fuzzy Systems **Text Book 1: chapter 13**	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Explore the contemporary optimization approaches	S.S.Rao	5 th edition,2019
2	Optimization techniques	A.K. Malik ,S.K. Yadav S.R. Yadav	2013

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Operations Research: An Introduction	Hamdy A Taha	10th Edition, 2017.
2	Operations Research Theory and Applications	J.K. Sharma	6th Edition,2016.

Course Outcomes:

Course outcome	Descriptions
CO1	**Understand** the principles and historical development of optimization
CO2	**Solve** linear programming problems using methods like simplex, graphical, and artificial variable techniques.
CO3	**Apply** optimization techniques to transportation and assignment problems
CO4	**Implement** unconstrained and constrained optimization methods
CO5	**Evaluate** advanced optimization algorithms like Genetic Algorithms

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3		1		1								
CO2	3	2	2	1		1							
CO3	3	1	2	1									
CO4		2											
CO5	1	2	1										

Department: CSE (Data Science)			Semester:	8
Subject: Project Work Phase – II				
Subject Code:	22DSPW803	L – T – P - S:	0-0-126-114	
			Credits:	8

Sl. No	Experiment Description
1	Student should give a presentation to the guide on the system requirements and design phase which includes hardware and software requirements, system architecture, high level design, low level design, system models, required implementation tools and algorithms.
2	Students are allowed to complete the project by implementing/simulating to generate expected results under the guidance of project guide.
3	Students are required to prepare the final report of the project with the generated results, future scope and conclusions.
4	Students have to demonstrate the working model of the project to their respective evaluation panel.
	Evaluation: ➢ Continuous evaluation is done by respective evaluation panel based on the regularity, technical knowledge and competence, programming skills, communication skills, collaborative learning, demonstration skills and documentation skills of the students. **Demonstration:** ➢ Students have to demonstrate the working model of the project to their respective project guide and for External examiner in the final exam.

Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** theoretical knowledge and engineering principles to design and implement a practical solution for a real-world problem.
CO2	Effectively **plan, manage, and execute** a multi-disciplinary project, adhering to timelines and resource constraints.
CO3	**Analyze, interpret, evaluate** experimental results and identify the areas for improvement.
CO4	**Communicate** complex technical information effectively through reports, presentations, and demonstrations.
CO5	**Demonstrate** ethical considerations, professional conduct, and an understanding of intellectual property rights through their team work.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	2	1		1	1	1	1		
CO2	2	2	3	1	2			1	2	2	2		
CO3	3	3	2	3	2			2	2	2			
CO4			2					3	2	3			
CO5								3	3	2	1		

Department: CSE (Data Science)			**Semester:**	**8**
Subject: Internship				
Subject Code:	**22DSIN804**	**L – T – P - S:**	**0-0-0-0**	
			Credits:	**0**

Sl. No	Internship Activity Description
1	**Course Overview:** •The Internship (Audit Course) is a **mandatory requirement** but carries **0 credits**. •There is **no Semester End Examination (SEE)**; evaluation is conducted solely through **Continuous Internal Evaluation (CIE)**.
2	Students must undertake an internship (Hybrid Mode) for a **minimum of 6 weeks and maximum of 8 weeks**, during the vacation between **4th and 5th semester** or **6th and 7th semester**.
3	Internships offered through **campus placements** (in Hybrid Mode) will also be considered valid for evaluation.
4	Students are required to submit a **detailed internship report** covering: internship organization, tasks performed, tools/technologies used, outcomes, challenges faced, and learning gained.
5	A **presentation** must be given by the student on their internship experience to the **Departmental Evaluation Committee**, which includes the faculty guide and a senior professor.
6	Students must appear for a **viva-voice examination** conducted by the evaluation panel, discussing technical and experiential aspects of the internship.

Evaluation Parameter	Marks
Assessment of Internship Relevancy by Departmental Committee	10
Company/Industry Assessment	10
Report Writing	10
Presentation	10
Viva-Voce	10
Total	**50**

7	**Evaluation : Continuous Internal Evaluation (CIE)** is based on: ➢ Relevance of the internship (assessed by faculty guide and senior professor) ➢ Assessment from the company/industry ➢ Quality and completeness of the report ➢ Presentation skills ➢ Viva-voce performance **CIE Marks Distribution (Total: 50 Marks)**

8	**Completion Criteria** ➢ **Grade card for 8th semester will be released only after successful completion** of internship. ➢ Minimum passing mark is **40% (PP Grade)**. ➢ If failed (NP Grade), the **entire internship must be repeated**.
9	**Demonstration:** Students are required to: ➢ Present their internship experience and outcomes to the **internal evaluation panel**. ➢ Submit proof of **internship completion certificate** from the company. ➢ Answer queries related to tools, technologies, and domain knowledge gained during the internship.

Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** theoretical engineering knowledge to real-world industry problems and practices.
CO2	**Demonstrate** problem-solving and technical skills in a professional work environment.
CO3	**Exhibit** professional ethics, teamwork, and interpersonal skills in a multidisciplinary setting.
CO4	**Communicate** effectively through technical reports, logs, and presentations based on internship
CO5	**Understand** industry practices, project management techniques, and business processes relevant to engineering.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	2	2						2		
CO2	2	3	3	3	3				3		2		
CO3			2			2	2	3	3	2			
CO4							1	2	3	3			
CO5			2		2	2	2		1	2	3		